

Passive target localization of target using TDOA measurements in the presence of NLOS using metaheuristic optimization algorithm

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Abstract

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Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

Location-based services have become essential in civilian and defense applications, including emergency response, logistics and warehousing, smart factories, autonomous robots, and context-aware healthcare. In many of these scenarios, especially surveillance and non-cooperative tracking, passive localization is attractive because the target does not need to transmit dedicated ranging signals and therefore remains energy efficient and difficult to detect [1–4].

Global Navigation Satellite Systems (GNSS), such as GPS and BeiDou, provide high-quality outdoor positioning, but their performance degrades severely indoors and in urban canyons because of signal attenuation, blockage, and multipath. Consequently, robust indoor positioning techniques are increasingly important for modern wireless systems and Internet-of-Things services [2,3]. Common indoor technologies include Wi-Fi, Bluetooth Low Energy, RFID, inertial sensing, and ultra-wideband (UWB), while the corresponding geometric ranging/positioning methods include angle of arrival (AOA), time of arrival (TOA), and time difference of arrival (TDOA) [1–3].

From a system perspective, localization methods are often classified as active or passive. Active methods usually require target-side transmissions or transponders, whereas passive methods estimate target position from received illuminator signals and their echoes. UWB is particularly important for both categories because its large bandwidth provides fine temporal resolution and enables high-accuracy delay estimation. Classical and modern

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studies have shown that TOA/TDOA-based UWB localization can achieve high precision under favorable propagation conditions [5–7].

However, indoor propagation is strongly affected by non-line-of-sight (NLOS) and multipath effects, which introduce excess delay and positive range bias, thereby degrading TOA/TDOA-based position estimation. This issue is well documented in both active and passive settings [8–11]. In the literature, NLOS errors are commonly modeled as additive non-negative biases in range (or delay) measurements, often with Gaussian, exponential, or mixture-based statistical assumptions depending on the environment [8–10]. Motivated by these challenges, this paper addresses passive target localization with TDOA measurements in mixed LOS/NLOS environments and focuses on improving robustness against NLOS-induced errors.

2. Localization problem

The proposed passive target localization problem uses TDOA signal parameter to estimate the passive target unknown position under LOS and NLOS propagation conditions. The considered two-dimensional localization system consists of a set of transmitters $\mathbf{x}_i^t = [x_i^t \ y_i^t]^T \in \mathbb{R}^2, \forall i \in \{1, 2, \dots, N\}$ with known coordinates, one passive receiver $\mathbf{x}_r = [x_r \ y_r]^T$, and one passive target at unknown position $\mathbf{x} = [x \ y]^T \in \mathbb{R}^2$. Without the loss of generality, the receiver is placed in the origin, i.e., $\mathbf{x}_r = [0 \ 0]^T$.

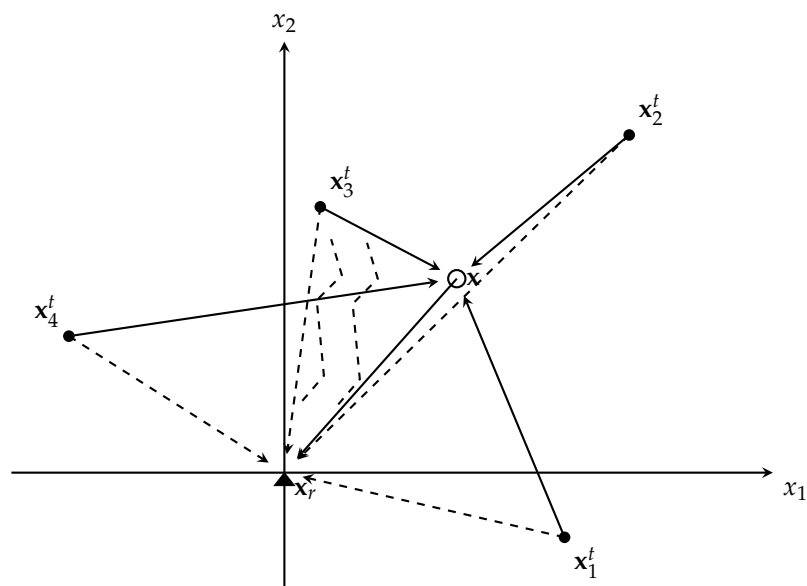


Figure 1. Geometric configuration of passive localization with transmitters $\mathbf{x}_i^t$, passive receiver $\mathbf{x}_r$, target $\mathbf{x}$, and representative LOS/NLOS propagation paths.

Assuming the time synchronization of transmitters and receiver is obtained, the passive receiver observes two signal paths for each transmitter: the direct path transmitter–receiver and the reflected path transmitter–target–receiver. The corresponding propagation distances are given by

$$d_{TR,i} = \|\mathbf{x}_i^t - \mathbf{x}_r\|_2, \quad d_{TTR,i} = \|\mathbf{x}_i^t - \mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_r\|_2. \quad (1)$$

The arrival times of direct and reflected paths are

$$t_{d,i} = \frac{d_{TR,i}}{c}, \quad t_{r,i} = \frac{d_{TTR,i}}{c}, \quad (2)$$

where c denotes propagation speed of light. Therefore, the noisy TDOA measurement is obtained as

$$t_i = t_{r,i} - t_{d,i}, \quad \forall i \in \{1, 2, \dots, N\}. \quad (3)$$

By multiplying with c , the range-difference measurement is

$$r_i = ct_i = (\|\mathbf{x}_i^t - \mathbf{x}\|_2 + \|\mathbf{x}_r - \mathbf{x}\|_2 - \|\mathbf{x}_i^t - \mathbf{x}_r\|_2) + n_i, \quad (4)$$

where n_i is zero-mean Gaussian measurement noise, i.e., $n_i \sim \mathcal{N}(0, \sigma^2)$. Since $\|\mathbf{x}_i^t - \mathbf{x}_r\|_2$ is known, we introduce

$$\tilde{r}_i = r_i + \|\mathbf{x}_i^t - \mathbf{x}_r\|_2, \quad (5)$$

which gives

$$\tilde{r}_i = \|\mathbf{x}_i^t - \mathbf{x}\|_2 + \|\mathbf{x}_r - \mathbf{x}\|_2 + n_i, \quad \forall i \in \{1, 2, \dots, N\}. \quad (6)$$

In realistic environments, some measurements are affected by NLOS propagation due to reflections, diffraction, and scattering. These effects produce additional positive range bias, which is modeled by term $b_i \geq 0$ [12]. Then the measurement model becomes

$$\tilde{r}_i = \|\mathbf{x}_i^t - \mathbf{x}\|_2 + \|\mathbf{x}_r - \mathbf{x}\|_2 + b_i + n_i. \quad (7)$$

To describe unknown LOS/NLOS condition of each measurement, binary variable z_i is introduced as

$$z_i = \begin{cases} 0, & \text{LOS} \\ 1, & \text{NLOS} \end{cases} \quad (8)$$

with probabilities

$$P(z_i = 0) = p_{LOS}, \quad P(z_i = 1) = p_{NLOS}, \quad (9)$$

where p_{LOS} denotes the probability that the i -th measurement is under LOS condition, and p_{NLOS} denotes the probability that the i -th measurement is under NLOS condition, such that they satisfy

$$p_{LOS} + p_{NLOS} = 1. \quad (10)$$

Under LOS condition, $b_i = 0$, while for NLOS condition the bias is modeled as a random variable with mean μ_b and variance σ_b^2 [8–10]. Therefore, the conditional measurement distribution can be represented by Gaussian mixture model

$$p(\tilde{r}_i|\mathbf{x}) = p_{LOS}\mathcal{N}(\tilde{r}_i; d_i(\mathbf{x}), \sigma^2) + p_{NLOS}\mathcal{N}(\tilde{r}_i; d_i(\mathbf{x}) + \mu_b, \sigma_b^2), \quad (11)$$

where

$$d_i(\mathbf{x}) = \|\mathbf{x}_i^t - \mathbf{x}\|_2 + \|\mathbf{x} - \mathbf{x}_r\|_2, \quad (12)$$

The appropriate vector form is defined as

$$\mathbf{d}(\mathbf{x}) = [d_1(\mathbf{x}), d_2(\mathbf{x}), \dots, d_N(\mathbf{x})]^T. \quad (13)$$

Hence, for the measurement vector $\tilde{\mathbf{r}} = [\tilde{r}_1, \tilde{r}_2, \dots, \tilde{r}_N]^T$, the corresponding vector-form likelihood can be written as

$$p(\tilde{\mathbf{r}}|\mathbf{x}) = \prod_{i=1}^N \left[p_{LOS}\mathcal{N}(\tilde{r}_i; d_i(\mathbf{x}), \sigma^2) + p_{NLOS}\mathcal{N}(\tilde{r}_i; d_i(\mathbf{x}) + \mu_b, \sigma_b^2) \right]. \quad (14)$$

Therefore, the target localization task is to estimate unknown target position by solving ML problem

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x}} p(\tilde{\mathbf{r}}|\mathbf{x}), \quad (15)$$

which is nonlinear and nonconvex due to Euclidean distance terms and NLOS mixture model.

3. Formulation of maximum likelihood estimator

Based on the model defined in Section 2, this section gives the compact likelihood form used for optimization.

Let residual term be defined as

$$e_i(\mathbf{x}) = \tilde{r}_i - d_i(\mathbf{x}), \quad \forall i \in \{1, 2, \dots, N\}. \quad (16)$$

Then the conditional measurement density can be written as a Gaussian mixture in residual form

$$p(\tilde{r}_i|\mathbf{x}) = p_{LOS}\mathcal{N}(e_i(\mathbf{x}); 0, \sigma^2) + p_{NLOS}\mathcal{N}(e_i(\mathbf{x}); \mu_b, \sigma_b^2). \quad (17)$$

Assuming conditional independence of measurements, joint likelihood becomes

$$L(\tilde{\mathbf{r}}|\mathbf{x}) = \prod_{i=1}^N \left[p_{LOS}\mathcal{N}(e_i(\mathbf{x}); 0, \sigma^2) + p_{NLOS}\mathcal{N}(e_i(\mathbf{x}); \mu_b, \sigma_b^2) \right]. \quad (18)$$

By substituting Gaussian density function, the full likelihood expression is obtained as

$$L(\tilde{\mathbf{r}}|\mathbf{x}) = \prod_{i=1}^N \left[p_{LOS} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(e_i(\mathbf{x}))^2}{2\sigma^2}\right) + p_{NLOS} \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(e_i(\mathbf{x}) - \mu_b)^2}{2\sigma_b^2}\right) \right]. \quad (19)$$

Accordingly, the log-likelihood function is

$$\ell(\mathbf{x}) = \ln L(\tilde{\mathbf{r}}|\mathbf{x}) = \sum_{i=1}^N \ln \left[p_{LOS}\mathcal{N}(e_i(\mathbf{x}); 0, \sigma^2) + p_{NLOS}\mathcal{N}(e_i(\mathbf{x}); \mu_b, \sigma_b^2) \right]. \quad (20)$$

The maximum likelihood estimate is obtained by

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x} \in \mathbb{R}^2} \ell(\mathbf{x}), \quad (21)$$

or equivalently

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{R}^2} (-\ell(\mathbf{x})). \quad (22)$$

For all-LOS case ($b_i = 0$), formulation reduces to nonlinear least-squares objective

$$J_{ML}(\mathbf{x}) = \sum_{i=1}^N (\tilde{r}_i - d_i(\mathbf{x}))^2, \quad (23)$$

with estimate

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{R}^2} J_{ML}(\mathbf{x}). \quad (24)$$

Due to nonlinear Euclidean distance terms and LOS/NLOS uncertainty, the resulting optimization problem is nonconvex and can be multimodal [5,11].

The objective function topology for the LOS case is shown in Figure 2.

Interactive 3D Plot of Negative Log Likelihood Function

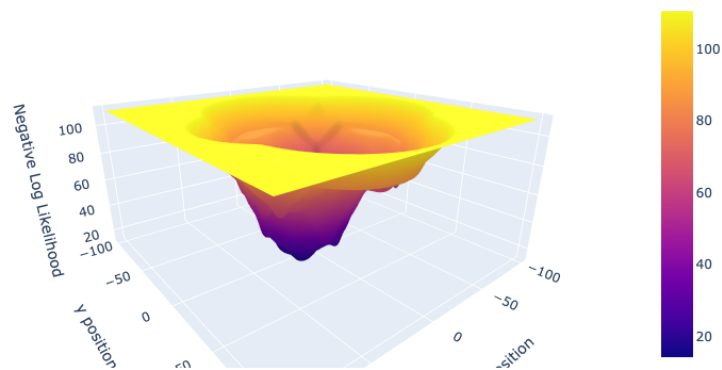


Figure 2. Objective function surface of the maximum likelihood estimator, $J_{ML}(\mathbf{x})$, illustrating its nonlinear and multimodal characteristics.

4. Differential Evolution for Position Estimation

To solve the nonlinear minimization in (22), we use differential evolution (DE), a population-based stochastic optimizer with strong performance on multimodal continuous problems [?]. In each generation, DE applies four operators: initialization, mutation, crossover, and selection.

4.1. Population Initialization

At generation G , the population is represented as

$$\mathbf{x}_i^{(G)} = [x_{i,1}^{(G)}, x_{i,2}^{(G)}, \dots, x_{i,n}^{(G)}]^T, \quad i = 1, \dots, N_P, \quad (25)$$

where n is the decision dimension and N_P is population size. Each component is sampled uniformly within the feasible bounds,

$$x_{i,j}^{(0)} = x_j^L + \text{rand}_{i,j} (x_j^U - x_j^L), \quad (26)$$

with $\text{rand}_{i,j} \sim \mathcal{U}(0,1)$.

4.2. Mutation Strategies

For each target vector $\mathbf{x}_i^{(G)}$, a mutant vector $\mathbf{v}_i^{(G+1)}$ is generated by scaled differences between randomly selected individuals. Typical schemes include [?]

$$\mathbf{v}_i^{(G+1)} = \mathbf{x}_{r_1}^{(G)} + F(\mathbf{x}_{r_2}^{(G)} - \mathbf{x}_{r_3}^{(G)}), \quad (27)$$

$$\mathbf{v}_i^{(G+1)} = \mathbf{x}_{best}^{(G)} + F(\mathbf{x}_{r_1}^{(G)} - \mathbf{x}_{r_2}^{(G)}), \quad (28)$$

and

$$\mathbf{v}_i^{(G+1)} = \mathbf{x}_i^{(G)} + F(\mathbf{x}_{best}^{(G)} - \mathbf{x}_i^{(G)}) + F(\mathbf{x}_{r_1}^{(G)} - \mathbf{x}_{r_2}^{(G)}), \quad (29)$$

where indices are mutually distinct and different from i . The scaling factor F controls mutation amplitude and directly affects convergence and robustness [?].

4.3. Binomial Crossover

The trial vector $\mathbf{u}_i^{(G+1)}$ mixes target and mutant vectors component-wise:

$$u_{i,j}^{(G+1)} = \begin{cases} v_{i,j}^{(G+1)} & \text{if } \text{rand}_{i,j} \leq CR \vee j = j_{\text{rand}} \\ x_{i,j}^{(G)} & \text{otherwise,} \end{cases} \quad (30)$$

where $CR \in [0, 1]$ is crossover rate and j_{rand} guarantees that at least one component is inherited from the mutant vector. Adaptive control of CR is often beneficial in practice [?].

4.4. Greedy Selection and Objective Evaluation

DE keeps the better individual between $\mathbf{x}_i^{(G)}$ and $\mathbf{u}_i^{(G+1)}$ using the objective value in (22):

$$\mathbf{x}_i^{(G+1)} = \begin{cases} \mathbf{u}_i^{(G+1)}, & \text{if } -\ell(\mathbf{u}_i^{(G+1)}) \leq -\ell(\mathbf{x}_i^{(G)}), \\ \mathbf{x}_i^{(G)}, & \text{otherwise.} \end{cases} \quad (31)$$

This one-to-one replacement rule preserves non-deterioration of the best-so-far solution and drives the search toward the maximum-likelihood estimate in challenging LOS/NLOS conditions.

5. Pufferfish Optimization Algorithm for Position Estimation

As an alternative to DE, we also consider the Pufferfish Optimization Algorithm (POA), a recent nature-inspired metaheuristic that models the interaction between a pufferfish and a predator [13]. POA is structured as a population-based iterative search with two complementary stages: an exploration stage associated with predator attack behavior and an exploitation stage associated with predator retreat after the pufferfish inflates its spiny body [13].

5.1. Representation and Initialization

Each candidate solution is a position vector in the search domain,

$$\mathbf{x}_i^{(G)} = [x_{i,1}^{(G)}, \dots, x_{i,n}^{(G)}]^T, \quad i = 1, \dots, N_P, \quad (32)$$

initialized uniformly in the feasible box

$$x_{i,j}^{(0)} = x_j^L + \rho_{i,j} (x_j^U - x_j^L), \quad \rho_{i,j} \sim \mathcal{U}(0, 1). \quad (33)$$

In our localization setup, $n = 2$ and the coordinates correspond to the unknown target position.

5.2. Two-Phase Search Mechanism

At each iteration, POA updates each candidate in two consecutive phases:

1. *Exploration phase*: a global movement step is applied to increase spatial diversity and probe unexplored regions.
2. *Exploitation phase*: a local refinement step is applied around promising regions to improve objective value.

After each phase, boundary handling is performed, and a greedy acceptance rule keeps the new position only when it improves fitness [13]. This mechanism supports a dynamic exploration–exploitation balance without adding many control parameters.

5.3. Objective Function and Selection Rule

For passive localization, each candidate $\mathbf{x}_i^{(G)}$ is evaluated by the negative log-likelihood objective in (22). The update can be written in compact form as

$$\mathbf{x}_i^{(G+1)} = \begin{cases} \mathbf{x}_{i,\text{new}}^{(G+1)}, & \text{if } -\ell(\mathbf{x}_{i,\text{new}}^{(G+1)}) \leq -\ell(\mathbf{x}_i^{(G)}), \\ \mathbf{x}_i^{(G)}, & \text{otherwise.} \end{cases} \quad (34)$$

The best candidate over all iterations is reported as the final estimate $\hat{\mathbf{x}}$.

5.4. Complexity and Practical Remarks

With population size N_p , decision dimension n , and maximum iterations T , the overall complexity is linear in these quantities, i.e., $\mathcal{O}(N_p n T)$, consistent with the original POA analysis [13]. For the present 2-D localization problem, this makes POA computationally tractable while remaining robust on nonconvex objective surfaces.

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References

1. Liu, H.; Darabi, H.; Banerjee, P.; Liu, J. Survey of Wireless Indoor Positioning Techniques and Systems. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* **2007**, *37*, 1067–1080. <https://doi.org/10.1109/TSMCC.2007.905750>.
2. Gu, Y.; Lo, A.; Niemegeers, I. A Survey of Indoor Positioning Systems for Wireless Personal Networks. *IEEE Communications Surveys & Tutorials* **2009**, *11*, 13–32. <https://doi.org/10.1109/SURV.2009.090103>.
3. Zafari, F.; Gkelias, A.; Leung, K.K. A Survey of Indoor Localization Systems and Technologies. *IEEE Communications Surveys & Tutorials* **2019**, *21*, 2568–2599. <https://doi.org/10.1109/COMST.2019.2911558>.
4. Shitvov, A.P.; Bialer, O.; Raviv, J.; Shmaliy, Y.S. Passive coherent location system positioning based on TDOA and FDOA. *Chinese Journal of Aeronautics* **2014**, *27*, 949–956. <https://doi.org/10.1016/j.cja.2014.06.004>.
5. Gezici, S.; Tian, Z.; Giannakis, G.B.; Kobayashi, H.; Molisch, A.F.; Poor, H.V.; Sahinoglu, Z. Localization via Ultra-Wideband Radios: A Look at Positioning Aspects for Future Sensor Networks. *IEEE Signal Processing Magazine* **2005**, *22*, 70–84. <https://doi.org/10.1109/MSP.2005.1458289>.
6. Alarifi, A.; Al-Salman, A.; Alsaleh, M.; Alnafessah, A.; Al-Hadhrani, S.; Al-Ammar, M.; Al-Khalifa, H.S. Ultra Wideband Indoor Positioning Technologies: Analysis and Recent Advances. *Sensors* **2016**, *16*, 707. <https://doi.org/10.3390/s16050707>.
7. Chan, Y.T.; Ho, K.C. A Simple and Efficient Estimator for Hyperbolic Location. *IEEE Transactions on Signal Processing* **1994**, *42*, 1905–1915. <https://doi.org/10.1109/78.301830>.
8. Chen, P.C. A Non-Line-of-Sight Error Mitigation Algorithm in Location Estimation. In Proceedings of the Proceedings of IEEE Wireless Communications and Networking Conference (WCNC), 1999, Vol. 1, pp. 316–320. <https://doi.org/10.1109/WCNC.1999.797838>.
9. Chan, Y.T.; Tsui, W.Y.; So, H.C.; Ching, P.C. Time-of-Arrival Based Localization Under NLOS Conditions. *IEEE Transactions on Vehicular Technology* **2006**, *55*, 17–24. <https://doi.org/10.1109/TVT.2005.861208>.
10. G"uvenc, I.; Chong, C.C. A Survey on TOA Based Wireless Localization and NLOS Mitigation Techniques. *IEEE Communications Surveys & Tutorials* **2009**, *11*, 107–124. <https://doi.org/10.1109/SURV.2009.090308>.
11. Prlja, M.; Milić, D.; Simić, M.; Majić, A.; Ljubičić, N. Passive Localization with TDOA Measurements in the Presence of NLOS Propagation. *Electronics* **2018**, *7*, 362. <https://doi.org/10.3390/electronics7120362>.
12. Angarano, S.; Mazzia, V.; Salvetti, F.; Fantin, G.; Chiaberge, M. Robust ultra-wideband range error mitigation with deep learning at the edge. *Engineering Applications of Artificial Intelligence* **2021**, *102*, 104278. <https://doi.org/10.1016/j.engappai.2021.104278>.
13. Al-Baik, O.; Alomari, S.; Alssayed, O.; Gochhait, S.; Leonova, I.; Dutta, U.; Malik, O.P.; Montazeri, Z.; Dehghani, M. Pufferfish Optimization Algorithm: A New Bio-Inspired Metaheuristic Algorithm for Solving Optimization Problems. *Biomimetics* **2024**, *9*, 65. <https://doi.org/10.3390/biomimetics9020065>.

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